^{36}Ar(pol d,^{3}He) 1993Ma50

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

1993Ma50: a 52-MeV polarized deuteron beam at 15 nA was produced from the Karlsruhe isochronous cyclotron at the Max-Plank-Institut fur Kernphysik. The target was 350-mbar 99.86% enriched ^{36}Ar gas. Reaction products were detected using a Si detector telescope consisting of a 250−μm ΔE-strip detector and a 1.5 -mm surface-barrier E detector with FWHM=130 keV. Measured angular distributions of cross sections $\sigma(E(^{3}He),\theta)$ and angular distributions of analyzing powers $iT_{11}(E(^{3}He),\theta)$. Deduced levels, J, π, L-transfers, and spectroscopic factors from DWBA analysis of $\sigma(E(^{3}He),\theta)$ and $iT_{11}(E(^{3}He),\theta)$.

1993Ma50 states that spectroscopic factors were obtained by fitting the forward-angle maxima. The agreement with the analyzing powers is only qualitative. The determination of spins and parities of final states is mainly based on a comparison with measured angular distributions leading to final states with known spins. The comparison with DWBA predictions serves as additional check. The data were taken only in a relatively small angle interval. This is sufficient for the spin determination but not favorable for the measurement of spectroscopic factors.

^{35}Cl Levels

Spectroscopic factor $C^{2}S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=2.95 is a normalization factor used by 1993Ma50 from 1966Ba54.

E(level)	J^{π}@	L@	$C^{2}S$@	Comments
0	3/2+†	2	1.72†	
1213 *4*	1/2+	0	0.78	J^{π}: $2s_{1/2}$.
1785# *50*			0.04	J^{π}=(5/2+) is given in Table 2 of 1993Ma50.
2676# *7*			0.16	E(level): possible doublet of 2645.6 and 2693.6 (1990En08). J^{π}=(3/2+) is given in Table 2 of 1993Ma50.
3002 *6*	5/2+‡	2	1.32‡	
3159 *20*	7/2−	3	0.19	J^{π}: $1f_{7/2}$.
3948 *40*	(3/2+)†	2	0.03†	E(level): possible doublet of 3918.48 and 3967.5 (1990En08). $C^{2}S$: for J^{π}=3/2+; 0.08 for J^{π}=1/2+ (1993Ma50).
4839# *12*				E(level): possible doublet of 4839.09 (1/2+,3/2) (1990En08) and an unspecified (1/2+) component. J^{π}=(5/2+) and (1/2+) are given in Table 2 of 1993Ma50. $C^{2}S$: 0.08 for J^{π}=5/2+; 0.03 for J^{π}=1/2+ (1993Ma50).
5155 *11*	(5/2+)‡	2	0.12‡	E(level): possible doublet of 5163.31 and 5215.8 (1990En08). J^{π}: (7/2−) and 5/2+ are given in Table 2 of 1993Ma50.
5583 *13*	5/2+‡	2	0.38‡	
5720 *11*	5/2+‡	2	1.10‡	
6136 *8*	5/2+‡	2	0.63‡	
6745 *12*	1/2+	0	0.09	J^{π}: $2s_{1/2}$. 5/2+ and 1/2+ are given in Table 2 of 1993Ma50. $C^{2}S$: for J^{π}=1/2+; 0.28 for J^{π}=5/2+ (1993Ma50).
6953 *43*	5/2+‡	2	0.49‡	
7181 *60*	5/2+‡	2	0.20‡	
7565 *20*	5/2+‡	2	0.30‡	
8007 *36*	1/2−	1	0.16	J^{π}: $1p_{1/2}$. 5/2+ and 1/2+ are given in Table 2 of 1993Ma50. $C^{2}S$: for J^{π}=1/2−; 0.20 for J^{π}=5/2+ (1993Ma50).
8204 *34*	5/2+‡	2	0.14‡	
8580# *20*				J^{π}=5/2+ is given in Table 2 of 1993Ma50. $C^{2}S$: 0.22 for J^{π}=5/2+ (1993Ma50).
8921# *70*				J^{π}=5/2+ is given in Table 2 of 1993Ma50. $C^{2}S$: 0.18 for J^{π}=5/2+ (1993Ma50).
9.22×10^{3}# *12*				J^{π}=5/2+ is given in Table 2 of 1993Ma50. $C^{2}S$: 0.12 for J^{π}=5/2+ (1993Ma50).
9514 *20*	1/2−		0.18	J^{π}: $1p_{1/2}$. 5/2+ and 1/2− are given in Table 2 of 1993Ma50.

Continued on next page (footnotes at end of table)

^{36}Ar(pol d,^{3}He) 1993Ma50 (continued)

^{35}Cl Levels (continued)

E(level)	J$^{\pi}$ @	C^2S @	Comments
			C^2S: for J^{π}=1/2$^-$; 0.30 for J^{π}=5/2$^+$ (1993Ma50).
9.82×10^3 *12*	1/2$^-$	0.10	J$^{\pi}$: 1p$_{1/2}$. 5/2$^+$ and 1/2$^-$ are given in Table 2 of 1993Ma50. C^2S: for J^{π}=1/2$^-$; 0.26 for J^{π}=5/2$^+$ (1993Ma50).
1.25×10^4			1d$_{5/2}$ is given for a range of 12.0-13.0 MeV in Fig. 4 of 1993Ma50.

† L-1/2 transfer from analyzing power measurements; 1d$_{3/2}$ proton transfer assumed in DWBA calculations.

‡ L+1/2 transfer from analyzing power measurements; 1d$_{5/2}$ proton transfer assumed in DWBA calculations.

No data on σ(E(^{3}He),θ), iT$_{11}$(E(^{3}He),θ), or L-transfer is given.

@ From DWBA analysis of the measured $\sigma(\theta)$ in 1993Ma50.